A Dose of Reality: Understanding the Origin of the Theory/Practice Dichotomy in Teacher Education from the Students' Point of View

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Research on preservice teachers' beliefs has aimed at increasing our understanding of students' prior conceptions in an effort to assess the impact on, and consequently inform the design of, teacher education programs. Many of these studies have reaffirmed the notion that teacher education programs seem to have little influence on the students' preconceived belief systems (Britzman, 1986; Goodman, 1988; Hollingsworth, 1989; Lacey, 1977; Lortie, 1975; McNeil, 1986; Zeichner, & Gore, 1990; Zeichner, & Tabanick, 1981). On the other hand, several studies have refuted this claim and demonstrated that students indeed have been significantly influenced by their academic course work. Alverman (1981) states that some students welcomed the inner struggle produced by the dissonance between the university courses and the practicum experience and that, in fact, it served as a driving force for encouraging reflection on the value of their teaching strategies and beliefs. Similarly, Grossman and Richert (1988) found that student teachers' conceptions on subject matter, general pedagogical knowledge, and awareness of pupil understanding were positively influenced by the university's professional course work.

In spite of their conflicting views, researchers on teachers' beliefs agree on the fundamental goal that teacher education programs must be responsive to the student teachers' prior beliefs in order to increase the impact of their professional preparation and broaden the knowledge base for teachers. However, researchers again become polarized by their distinct visions of how this common goal should be accomplished. For example, Grossman and Richert (1988) suggest that it may be premature to introduce student teachers to issues related to pupils' diverse abilities for understanding before the students gain school experience. They add that once students have experience in the field their capacity for understanding student diversity in abilities and their need for understanding how to manage it grows rapidly (p. 61). Hollingsworth (1989) agrees. Her research team interviewed and observed 14 elementary and secondary preservice teachers throughout their professional education and found that the students' understanding of how to respond to pupils' diverse reading skills was just as diverse. She concluded that even though these students were exposed to 15 hours of lecture/discussion sessions on reading during the course of their program, some students seemed to have forgotten how to use this information. Hence, Hollingsworth explained that, based on findings such as this, it may be necessary to *reverse the order of program content so that the information on remediation and assessment of problem readers could occur in the second semester when preservice teachers had more experience with both the content and pedagogy of reading and were placed in classrooms with students who have more learning difficulties. (p. 187)*

This statement is in direct contrast with Goodman (1988) who proposes a more proactive approach to teacher education by first, and foremost, identifying the students' intuitive screens, that is, the points of reference students use to make sense of their experiences as they sift through the information (educational theories, ideas and strategies) presented to them during their teacher education (p. 134). Goodman explains that once the students' screens have been compiled, specific *interventions can be devised to reinforce certain conceptions, while in another instance . . . find ways to help a student alter his or her initial perspectives* (p. 134). So, in addition to the difficult task of deciding what topic should be offered to students before or after they have had practicum experience, some researchers suggest that strategies should also be devised to alter the students' prior conceptions. This elicits

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many serious questions on what the goals of teacher education programs should be and to what extent the end should justify the means.

Disturbing Propositions

In recent years, several educators have investigated teachers' perspectives relating to action. Research on the repertoire of metaphorical figures teachers use to describe their experiences has been very useful in unpacking the meaning teachers construct from their professional reality as they see it (Bullough, 1991; Marshall, 1990; Munby, 1986; Munby, & Russell, 1990; Tobin, 1990). However, we must ask what purpose is served by learning about the students' conceptions of teaching and learning in their own language. Is what teachers represent in their espoused metaphors what they actually do in the classroom?

Tobin (1990) best illustrates the use of metaphors in the study of teachers' thinking: *By conceptualizing teachers' beliefs and roles through the metaphors they use, and then introducing other, more appropriate metaphors, teacher change can be implemented* (p. 127). Hewson and Hewson (1987a, 1987b, 1988) agree with this notion. They add that, in the case of science teacher education, specific activities should be incorporated to enhance the formation of appropriate conceptions of science. Hewson and Hewson (1987b) believe this can be accomplished by using a conceptual change model of learning (CCM), that is, learning as a process which may involve changing a person's conceptions in addition to adding new knowledge to what is already there (p. 427).

This suggestion for teacher education programs presupposes that student teachers will accept what educators perceive as the appropriate metaphors. It is disturbing to think that the students' own metaphors could be used to manipulate their perceptions instead of using metaphors as heuristic devices to help them reflect on what is meaningful and relevant to them, and on what aspects of their professional education they themselves see in need of improvement. Tobin (1990), Munby and Russell (1990) and others suggest that as student teachers closely examine the metaphors they use in learning to teach, they will become aware of whether conflicts exist between their espoused metaphors and the various roles and beliefs emerging from their classroom experiences. This may be the fundamental role of the study of metaphors used by teachers in professional programs. This role is also congruent with a constructivist perspective that respects the students' experiences and ways of making sense of their world. However, a teacher education program that claims a constructivist perspective and the study of the students' metaphors as part of their professional preparation, but then introduces what the educators consider to be the appropriate metaphors is intellectually dishonest. Again, what is the main goal? Is it to devise effective tools for uncritical indoctrination or to engage students in meaningful explorations of educational theories?

This brings us to the second question posed above: Are we to believe that an individual's espoused metaphor will always translate into action? Individuals may have metaphors that reflect their beliefs and desired actions, but a disposition to act is not the same as acting on a disposition, an observation with important ramifications for those who seek to incorporate the analysis of students' metaphors as part of their professional preparation. There is great value in helping student teachers become aware of their own beliefs on teaching and learning and of the metaphors they use to represent their desire for action. There is also a strong indication that this process may aid students in recognizing where their metaphors may conflict with the appropriate ways of teaching and the appropriate ways pupils learn. Yet awareness of conflicts does not necessarily translate into action or what some educators consider to be the expected action. It is possible that students may lack the tools to resolve the perceived conflicts among their own beliefs, the philosophy of the teacher education program they are pursuing, and the appealing jingoism of other approaches to teaching and learning espoused in the literature.

These and other studies offer a variety of insightful, challenging and appealing suggestions for improving teacher education. In this paper I argue that some of these researchers appear to have engaged in a language of persuasion and might have lost sight of the very people they attempt to influence. In other words, some researchers appear originally concerned about increasing the understanding of the student teachers' prior beliefs, but to what end? Is it to persuade them into accepting

program-specific education theories, values and beliefs? Every teacher education program explicitly or implicitly seeks to do just that for many reasons, such as to protect the pupils' interests, follow some professional canons for certification, or contribute to the betterment of society. Regardless of the reasons or good intentions, some investigators have overlooked the essence of their fundamental goal to enhance the understanding of the students' belief systems before attempting to alter their conceptions. I propose that to understand better the intuitive screens student teachers bring into teacher education programs, we must first pay close attention to the language of perception that students use to attach meaning to their experiences in two different contexts, the university course work and the practicum component. To discern the student teachers' prior beliefs and how these beliefs are affected as students progress through their programs, we must first be aware of the distinctions students construct between what they consider practical versus theoretical, or ideal versus real in their professional preparation. In this paper I report on the preliminary findings from a year-long study involving six science student teachers. I will contrast my analysis with the findings of other researchers to add to the debate on the pervasive dichotomy the students themselves perceive between theory and practice as they undertake the complex process of learning to be teachers.

Method

Site selection and methodology

At the beginning of their university course work, I invited six science student teachers (five men and one woman) to participate in this project on the basis of their willingness to contribute to the study, their chosen geographical location (same school district), and the diversity of their subject-matter specialization (all of the secondary sciences were represented). These students were originally assigned to three different schools by a placement officer.

I used an ethnographic approach for data collection and analysis. I was a participant observer, acting occasionally as a resource person for the students, an experienced teacher available for informal consultation with them at any time. In addition, as a resource person, I had access to schools to gather field notes and to observe and videotape some of the students' lessons. I was also able to observe students in their university classes. It was important to make clear to students that as a resource person I was non-evaluative and non-judgmental. My comments were not going to influence their standing on the practicum or on their academic courses, since I was to keep their views confidential.

The Program

Students at this university must hold a bachelor's degree in their subject area specialization before admission to the full (12-month), intensive teacher preparation program. This program is divided into three sections. The first is a professional semester in which students attend education courses full-time from September to December. This term is interrupted by a two-week orientation practicum from late October to mid-November. During this practicum students mostly observe in the classroom and engage in limited teaching. They are usually placed in the same school for the extended, 13-week practicum. The extended practicum is the bulk of the students' second term which runs from January to April. For the last term, running from May to August, students again devote full-time to foundations of education courses.

Data Collection

I used multiple data sources to inform this analysis. Each student participated in three in-depth interviews: one before the orientation practicum (Interview A, after five weeks of course work), one after the short practicum (Interview B), and one at mid-point of the extended practicum (Interview C, after 7-8 weeks into the extended practicum). In order to stimulate conversation on the students' beliefs during our first interview, I gave them a sheet with graphic representations of six commonly held metaphors on teaching and learning: a person working with clay, a guide and a traveller going for a hike, a gas attendant filling up a car with fuel, construction workers at a construction site, a child throwing pebbles into a pond, and a gardener tending a garden. I then encouraged students to choose the metaphor(s) that best represented their views or to provide a metaphor that they felt was more appropriate. I drew ques-

tions for the interviews from field observations, in-class videotapes, numerous informal conversations, and the students' Views on Learning and Teaching (VOLAT) books given to them at the beginning of the term. I encouraged them to write down their reactions to any events, comments, or ideas that influenced their own views on teaching and learning. I fully transcribed all interviews and made them available to the participants.

Analysis and Interpretation

Following some of the suggestions outlined by Spradley (1979, 1980), Hammersley and Atkinson, (1983) and Goetz and LeCompte (1984) for the design and analysis of ethnographic studies, I reread and coded the data into specific categories. I then compared and contrasted these emerging themes or categories to crystallize the interpretation of the data. I subsequently organized highlights of the student teachers' explanations of critical incidents into profiles. The students read the profiles at each interview and reacted to my comments. This technique provided an opportunity for the students to further elaborate on issues relevant to them. In the same fashion, this methodological approach allowed me to continue probing the impact of both the university program and the school practicum on the students' beliefs about teaching and learning.

Drawing from these multiple data sources, I triangulated the data to corroborate their trustworthiness and the emergence of possible themes and categories. I pursued cases of negative evidence to verify the impact of some factors on the student teachers' evolving conceptions of teaching and learning. I paid special attention to the concept of discipline subjectivity (Erickson, 1973) by submitting all phases of the research to continuous and rigorous questioning and reevaluation.

Students' Belief Systems in the University Context (Academic Course Work)

During their teacher education programs, these students very early become wary of the sense of artificiality surrounding the process of learning to be a teacher. They attend some classes in crowded lecture halls, listen to educational theories and strategies that challenge the value of how they have been taught in the last 16 or so years, and then try to translate and implement what they have learned in borrowed classrooms during the school practicum. They do this because they want to be teachers, but they soon begin to demonstrate strong feelings of disappointment and disenchantment about their university education. This may very well be the initial warning signs that they are about to embark on a long and stressful journey marked by frustration, confusion and miscommunication. This certainly is not the program's goal, so what is the root of the students' predicaments? We may begin to understand this by looking at the students' conceptions on teaching and learning as they enter their teacher education programs.

In this study, base-line profiles of the student teachers' personal philosophies on teaching and learning science were constructed by discussing their reasons for going into teaching, role models (positive and negative), beliefs, expected barriers, and the most appropriate metaphor(s) that encapsulated their views. Four of the six students (4 of 6) selected the metaphor of the guide and the traveller to represent their views on teaching and learning science. While there was some variation in their explanations, they generally saw the teacher as the guide and the pupil as the traveller, with occasional reversal of roles to allow for teachers to learn from their interactions with the pupils. Students also commented that positive role models (high school teachers, university instructors, or parents who were teachers) were sources of inspiration of how they would like to be perceived as teachers. On this point, their comments centered on their relationship with students and how they would like to avoid becoming like the teachers with whom they had difficulties. For instance, Barton explained that his high school teacher had favourites and that he would pick on students and embarrass them in front of the class (Interview A). Ellen said that she admired some of her teachers because they respected the students, but she also had *a lot of teachers who basically got up, taught whatever, just treated* (pupils) *like a bunch of idiots* (Interview A).

Participants appeared to have well-defined personal theories of learning. For instance, Chris believed that the learner has to be willing to learn before learning can take place. *You can promote a person's willingness to learn, but you cannot force them to learn.* Frank also had an idiosyncratically constructivist perspective of learning. He defined

learning *as self-realization—all of a sudden, it clicks in your mind* (Interview A). He added that in his role as a teacher he can only guide and clarify information (Interview A), but learning was ultimately an individual act.

With their preconceived beliefs on what constitutes good teaching and learning in science, students also had well-articulated fears. Several of them were concerned about what they expected as barriers to professional growth during the practicum. Among these were Barton's preoccupation with getting too close to the students because he did not believe in the idea of people being above people, or having to play the disciplinarian role to maintain control (Interview A). Others were also concerned about how to establish a positive relationship with their sponsor teachers so that they could try new and creative things in the classroom. All of the student teachers perceived the extended practicum as the most important component of their professional education. Some demonstrated a great deal of anxiety about it. Ellen commented, *As a student teacher, I would have to—a certain degree—play the game. Do what they* [sponsor teacher and/or faculty advisor] *want me to do. You know, if they are not receptive toward me trying new things, then fine. I mean, it would upset me, probably make me mad, but there is nothing I can do about it, and I want to do well.* (Interview A)

These student teachers entered the teacher education program with a revealing awareness of their beliefs, not only about teaching and learning but about the possible barriers that may interfere in their professional growth. In light of this, did they feel the university had prepared them well before they began the practicum component? After five weeks of on-campus course work and just before they began their two-week orientation practicum, students recognized the usefulness of their courses but wished the content was more practical and less theoretical. Dean was frustrated because he could not understand how he was expected to learn about classroom management (his main preoccupation before the practicum) in only two 90-minute lectures in a crowded lecture hall along with 300 other students. He added that *If they could take* [classroom management] *and apply it directly rather than talking about it abstractly (Interview A)*, these lectures would make more sense to him. Other students were disappointed that their instructors did not model the teaching and learning theories they were espousing, *We're being taught about getting the student motivated and interested and do a variety of things like provide opportunities for quality learning and help them think for themselves, and then the way they teach us, it's almost completely opposite of what they tell us.* (Interview A with Alfred)

These findings suggest that the pervasive chasm between what constitutes theory and what constitutes practice for preservice teachers begins very early in their course work and grows from this point onward. Viewed from the students' point of view, this scenario suggests that students are definitely not empty vessels in which faculty can pour university-sanctioned educational theories and values. While the university speaks to students through the language of persuasion, student teachers listen with the language of perception. Five weeks into their program, students already spoke of being ready to play the game to survive the practicum experience, and of academic course work being too theoretical and unresponsive to their well-defined needs and expectations. Perhaps it is this perceived lack of response on the part of the university that aggravates rather than alleviates the students' conceptions that their course work is too theoretical and ideal and not as practical and real as they had expected. Researchers should investigate where the students get the impression that learning how to be a teacher can be packaged into easily transferable sets of classroom management routines, teaching strategies or learning-enhancing devices. Could it be that the university program is inadvertently sending these positivistic mixed messages to students, i.e, if they learn how to do A they can accomplish B?

In an effort to further pursue the reasons for the widening dichotomy students perceive between what they consider to be theoretical versus practical, I discuss below the impact of the practicum experience on their prior beliefs.

Students' Belief Systems in the School Context (Practicum)

I selected students' comments that follow mainly from two in-depth interviews, one right after their two-week orientation practicum (during the fall semester) and the other after the students completed about 8 weeks of their thirteen-week

extended practicum (in the winter semester). I also used numerous informal conversations, in-class observations (videotapes), and field notes to encourage discussions of the events the students found most critical.

Picking Tricks from a Bag

Students' feeling of frustration and disappointment increased considerably after their first attempts at teaching. They believed that their course work was somewhat misleading, since they could not easily transfer the teaching strategies learned at university into the school context. Barton recounts his first attempt at teaching grade 11 biology, *I spent the first week doing a lot of observing and helping in labs. Then [in the second week] I taught a lesson and it was like talking to silence. You know, like this is my first experience of getting up there. In all our lessons that we do out here [at university], in little seminars, you talk and people respond. You know, there's interaction. And then you get sort of believing that that's the way it's going to be in the classroom. You're standing up there trying to get simple points out of [pupils] and they're not talking because they're not used to talking. And it's like dead silence. You're talking to silence. And it's also like 'whoa!' I felt that I was bombing. I felt that I was just biting it really badly so I cut it short at 45 minutes. And then I talked to the teacher after and he said I did fine. Right? He couldn't see any problems and stuff. That was weird because you're inside and you're looking out and you sort of get that tunnelling sound in your ears. And you're so distracted by the fact that you're not getting the results that you expected to get.* (Interview B, emphasis mine)

It is hard to imagine anyone having a perfect first lesson, but what is striking here is the sense of bewilderment and disappointment this student teacher felt when his well-prepared lesson and appealing visuals did not seem to excite his pupils as much as they did his colleagues in the university classroom. Student teachers get so used to talking about their subjects and teaching micro-lessons to fellow science students and interested professors that they go to schools with imprinted preconceptions of an all-too-different reality. The participants in this project give the impression that just a few warning words and concrete ideas about how to implement the same interesting lesson to a group of less receptive students may have been enough to prevent the apparent and sudden realization that the school context is very different from the university setting. Perhaps this action may have prevented Barton from concluding that everything in this program seems to be geared towards teaching the people who want to listen. We *are not really told how to deal with the kids that do not want to be there and have to be there* (Interview B).

Barton's comments are well echoed by other student teachers from several other studies, such as elementary and secondary reading students in Hollingsworth's (1989) work, secondary English and Social Studies student teachers in Britzman's (1991) book *Practice Makes Practice*, elementary preservice teachers in Goodman's (1988) study, as well as first-year teachers in Zeichner and Tabachnick's (1985) two-year study.

My conversations with the students make it clear that the short-practicum gave them many opportunities to deal with fundamental pedagogical issues, such as the diversity of students' learning abilities, difficulties in establishing and maintaining pupil motivation, the necessity to have effective time management, the variety of administrative functions of a teacher, and the discovery that many of their sponsor teachers taught in a manner opposed to the university program's philosophy. Nevertheless, the participants in this study gave their course work very little credit for allowing them to recognize these issues, and they hoped that during the extended practicum they would find a better correlation between the theory (their university course) and practice (the practicum). I will discuss some of these issues in more detail in the next section. What should be noted at this point is that the student teachers maintained the expectation of being provided with fail-safe tools by the university. Hence, they seemed to walk in their classrooms with an imaginary bag of tricks, full of ideas, strategies and educational theories of teaching and learning, but they had difficulties figuring out which trick to use and under what conditions. When they felt that the selected trick did not work, or did not work as well as it did in university, they felt frustrated and betrayed. Dean articulated these views, *There's so much coming at me, I don't know which way to go, which to modify, what works, what doesn't work. And then I think, in the process and in that confusion, sometimes I drop some of the things that*

worked. So that's the state of confusion that I'm seeing now. I'm just trying to keep up with all of this growth. (Interview C)

This misconception on the part of the students that educational principles are but tricks may be at the heart of their bewilderment when they perceive their academic course work as largely devoid of relevance to what occurs in schools. This problem may be compounded by the fact that the university may be sending mixed messages to the students by avoiding to directly tie the educational concepts students are expected to learn with concrete school case scenarios. If students enter and leave the practicum thinking that they are learning a series of tricks that can be easily replicated from the academic context to the school context, then they are only superficially understanding the theories of teaching and learning they are exposed to.

In light of the limitations student teachers perceive so far, what do they actually do as they progress though the extended practicum? How are their conceptions of teaching and learning affected under these conditions?

Expanding the Bag of Tricks

Even at the mid-point of their extended practicum, and in spite of the dissonance students perceived between their academic course work and their practicum experience, all students held their previously selected metaphor. The participants, however, readjusted their actions to fit, as some of them call it, the real world, and they saw the sponsor teacher as an important technical advisor to make that happen. Barton elaborated, *What I am getting from (my sponsor teacher) is a lot of how it actually is done. In some cases you can only do so much. He's really helpful. So, he has sort of changed what I believe is real as far as applying it in the short term.* (Interview C)

Some students saw the faculty advisor as an emissary of theories and ideas even more distant from the reality of the classroom to some students. Dean felt very pressured by the curriculum, time constraints and his sponsor teacher to cover the subject matter. Thus he had much difficulty translating what he learned at university (what he saw as theoretical and ideal) to what he could do in the classroom (what he saw as practical and real). *The faculty advisor wants me to try and incorporate some the of things we learned in junior science (methods class). And that's what I'm planning on doing is trying to incorporate them. And I put in, little tidbits here and there as much as I could fit in. It's just that I'm finding that it's difficult to apply those ideas because they're sort of framed in a theoretical framework and I have to translate that theoretical framework into a practical, realistic hands-on application and that takes a bit of work.* (Interview C with Dean)

This situational adjustment is what Lacey (1977) refers to as strategic compliance, or when an individual complies with the authority figure's definition of the situation and the constraints of the situation but retains private reservations about them (p. 72). It is important for researchers not to confuse strategic compliance as an act of capitulation by which students teachers surrender their ability and desire to resist institutional or program constrains. Elsewhere (Rodriguez, 1992a), I argue that when researchers mainly report on the participants' negative experiences, they might ignore the student teachers' covert and overt strategies for making their voices heard and acting on their own agendas. Ellen, for example, saw potential barriers for acting on her beliefs about teaching and learning *not like a stone wall that you are never get across it, but as a picket fence that you can pull boards off once in a while* (Interview A). Dean also saw this strategic compliance as a necessary, but transitory, readjustment and that the situation would be much different when he became a real teacher, *Quality learning has fallen by the wayside because of trying to deal with these much bigger issues. The quality of learning is sort of the icing or the frill to me right now, at this point. I think when I start teaching it will be more important. Right now, I just find that I have a lot of variables to contend with. And minimizing those variables is sort of survival.* (Interview C)

The chasm between what some students perceived as theory and practice widened and they felt unprepared to build a bridge between the two without the tools. As in the short practicum, they had well-articulated ideas about how their academic preparation could have been more relevant or practical before they began the practicum. In their eyes, the science methods courses—where they had opportunities to try demonstrations, labs, and peer teaching—were considered practical courses. This is because they felt they were actually translating

some of the teaching strategies and principles of learning from their notes to an actual situation. The foundations courses (which cover topics on educational psychology and theories of education) were found to have little or no impact on what they did in the classroom. Dean, one of the students who had the most difficulty during the practicum, explained his vision of a more practical and applicable academic component, *I think it would have been great if someone could have talked to us or we could have had a chance to hash over the idea of maintaining the pace [of instruction] and dealing with the content, the curriculum, time constraints and quality learning. All four—all of these [topics] at the same time. Because, really, we talked about quality of learning and we had this ideal class, you know, everyone wants to learn. Every kid in the classroom is an ideal kid. They're academically oriented, their behavior is reasonable, you can deal with them using these techniques. If you do A, then B will follow. It just doesn't work that way. And I think we need a lot more of the practical.* (Interview C)

All six students commented on the same point: they all would like to see and discuss more practical situations in their university courses. Among the many suggestions for improving the academic preparation of teachers were the discussion of videotaped school scenarios where a teacher was resolving a conflict with a pupil, teachers using various instructional techniques to teach complex topics to receptive and unreceptive pupils, and role-playing school scenarios at the university. These are good suggestions, but educators and researchers must be cautious. Are students really asking that their already crowded bag of tricks be expanded? This might aggravate their abilities to decide which trick to use and for which pupils. Student teachers may indeed need more practical connections between their academic preparation and life at schools, but what they appear to most need is guidance in learning the essence of educational theories as they relate to practice. For example, a teacher education program based on a constructivist theory of learning and a philosophy of reflective practice would have little to add to the professional preparation of their students by giving them more practical examples of what to do in the classroom without the students critically understanding why they should use those strategies. The students in this study became quite frustrated when a well-rehearsed, well-replicated demonstration or lesson did not work out as well in the practicum as it did at the university.

Implications for Teacher Education

All the participants in this study maintained their espoused personal philosophy of teaching and learning and the same metaphor(s) they originally selected. Their views were not rigid, however. They adjusted and readjusted their perspectives to fit into the restraints encountered in the school context. Nevertheless, they did not seem to lose the essence of their prior beliefs about teaching and learning. This finding supports Britzman's (1986) claim that the *tabula rasa* approach to teacher education is not effective. As found in this study, and as suggested by Gunstone and his associates (1989), teacher education programs must develop ways to tap into the students' belief systems very early in the program in order to assist students in constructing more significant meaning and deeper understanding of the educational theories presented to them. This may not be effectively accomplished by simply rearranging the order of pedagogical topics offered to students before or after their practicum experience. It may also be of little help to become more aware of the students' prior beliefs if the intention is to persuade them into uncritically accepting established program goals and values.

It seems, then, that before educators initiate changes into existing programs, or make attempts to alter the students' beliefs, they should gain a better understanding of the reasons why students appear to perceive their academic preparation as theoretical and ideal and the practicum as the only source of practical and real experiences. Teacher education programs may begin an effective dialogue with students by abandoning the language of persuasion and tuning into the students' language of perception. The results of this study suggest that it may be more productive to work with the strong conceptions of learning and teaching students bring with them into the program, acknowledge their value, and move to provide concrete opportunities for students to experience alternate strategies and theories. According to the participants in this project, opportunities to experience specific educational concepts and pedagogical strategies would

involve activities such as critical analysis of videotaped case studies, peer teaching, role playing, and group discussions before and after their practicum. This approach may assist students in constructing a bridge between theory and practice without dismissing their 16 or more years of experience in the classroom, or as Lortie (1975) suggests apprenticeship of observation on how to teach.

The results discussed herein correspond only to a preliminary analysis of an ongoing project. What would have happened to the students' conceptions by the time they had completed their teacher preparation program? Would they still have held their views that practice and theory are two separate entities which they had felt unprepared to bridge, or would they have found relevant connections? Would their last-term course work have influenced their views after completing the school practicum. Possible answers to these questions may arise when this project is concluded. The impact of several other factors on preservice teachers' conceptions not discussed here due to time and space constraints are also being investigated, such as the influence of the sponsor teacher, the faculty advisor, the pupils, and the curriculum. How student teachers negotiate meaning from their experiences and how they translate these meanings into action may be the focal points for increasing our understanding of how teacher education programs may be inadvertently widening the gap between theory and practice from the students' point of view. Consequently, educators may then be better prepared to design and implement more significant and efficacious professional programs.

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